Bayslope

ANALYSIS REPORT

US9786153B2 VS. SAMSUNG SMART TAG 2

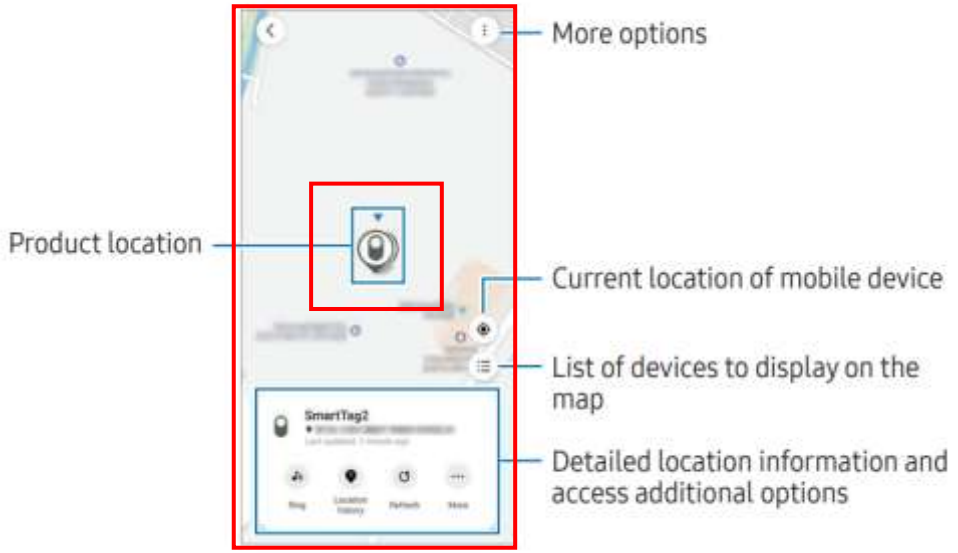
Privileged and Highly Confidential

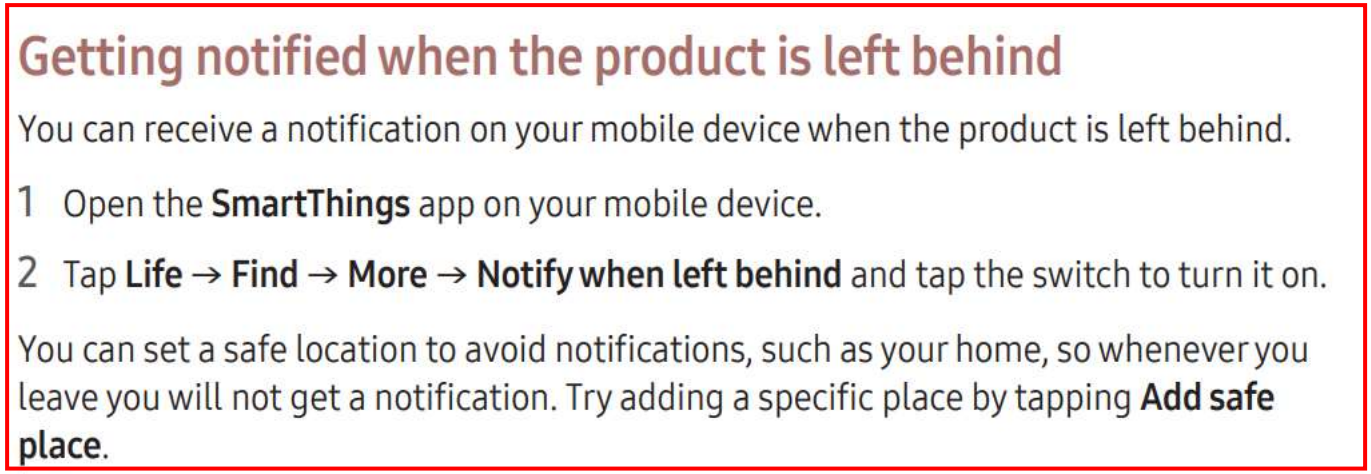
SUBJECT PATENT	US9786153B2
PRIORITY DATE	NOV 26, 2014
PATENT TECHNOLOGY	ASSET TRACKING | LOCATION TRACKING
CHART STRENGTH	HIGH
ENTITY NAME	SAMSUNG
PRODUCT NAME	SAMSUNG SMART TAG 2
SAMSUNG REVENUE	$206 BILLION (SOURCE)
PRODUCT CATEGORY	LOCATION TRACKER | DEVICE | APP
BUSINESS TYPE	B2C | BUSINESS TO CUSTOMER
PRODUCT RELEASE DATE	OCTOBER 2023 (SOURCE)

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17. A method of transmitting an alert comprising:	"Referring now to the drawings, FIGS. 1-4 illustrate control devices, according to one or more embodiments disclosed herein. Control devices are configured to receive from and send to a tagged receiver (as shown in FIGS. 5A-15) signals indicative of tracking, locating, or alerting a status of the asset to which the tagged receiver is attached. For example, a user may use the control device to locate an asset with a tagged	**Comment**: *A method of transmitting an alert (i.e., Samsung Smart Tag 2 sends alert messages to the user).* *A SmartTag 2 sends alert or notification to the user through registered mobile device.*  Source: https://www.samsung.com/us/mobile/mobile-accessories/phones/galaxy-smarttag2-black-ei-t5600bbegus/

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	receiver attached thereto on a map. In other embodiments, the control device may remotely activate an alert of the tagged receiver. Control devices may be kept in a variety of locations but may be small enough to carry with a user." *[I:41-36; col: 6]*	**Getting notified when the product is left behind** You can receive a notification on your mobile device when the product is left behind. 1 Open the **SmartThings** app on your mobile device. 2 Tap **Life** → **Find** → **More** → **Notify when left behind** and tap the switch to turn it on. You can set a safe location to avoid notifications, such as your home, so whenever you leave you will not get a notification. Try adding a specific place by tapping **Add safe place**. Source: https://www.bhphotovideo.com/lit_files/1056749.pdf
syncing a tagged receiver configured to attach to and track an asset to a control device such that the tagged receiver and the control device	"In other embodiments, both the control device 200 and the tagged receiver 300 may be Bluetooth enabled and sync using a Bluetooth pairing process. However, it	**Comment**: *Syncing a tagged receiver (i.e., SmartTag 2) configured to attach to and track an asset (i.e., personal items such as bags, keys etc) to a control device (i.e., mobile phone) such that the tagged receiver and the control device wirelessly communicate with each other.* *Samsung Galaxy SmartTag 2 can be attached to personal items like bags, keys, or bikes to track their location. A smartphone, running the SmartThings app, is used to set up, monitor, and control the SmartTag 2. The synchronisation between the SmartTag 2 and the smartphone occurs during the pairing process via Bluetooth Low Energy (BLE)/Wi-Fi, depending on the supported device.*

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wirelessly communicate with each other;	should be understood that in other embodiments the control device 200 may be synced with the tagged receiver 300 in any other manner that enables the control device 200 and the tagged receiver 300 to communicate via network 222. Once the control device 200 and the tagged receiver 300 are synced, the tagged receiver 300, or the asset attached thereto, may be displayed on the display 224 of the control device 200.	**About Galaxy SmartTag2** The Galaxy SmartTag2 is a product that can be attached to pets or objects in order to prevent their loss and to stay updated on their location. You can also launch automation features through the product using its button. Connect a product to your mobile device through the SmartThings app to check the location of the product and get information about where the product was lost. Source: https://www.bhphotovideo.com/lit_files/1056749.pdf **Connecting products to a mobile device** 1 Open the **SmartThings** app on your mobile device. Update the SmartThings app to the latest version. 2 Remove the film between the battery tray and the battery, or insert the battery. Alternatively, press the product's button. The product will make a sound and be activated. 3 When the pop-up window informing that your product is detected appears on your mobile device, tap **Add**. If the connection pop-up window does not appear, tap **Devices** → + . Tap **Add** under **Samsung devices** and tap **Tag/Tracker**. 4 Follow the on-screen instructions to complete the connection. When the connection is complete, the tag's current location will be displayed on the **SmartThings Find** map and the product will appear on the list of devices in the SmartThings app.

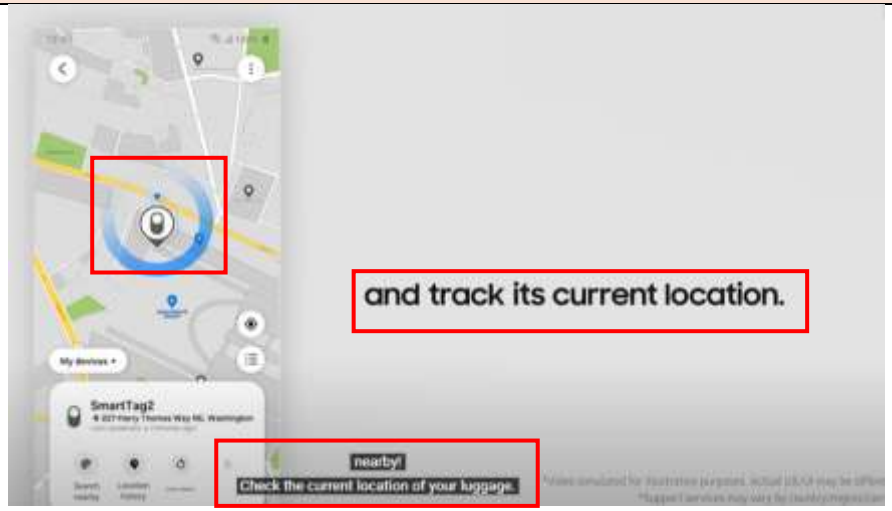
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	While syncing the control device 200 and the tagged receiver 300 is depicted as the first step in the flowchart of FIG. 19, in other embodiments, the link may be established later.*"[I:27-31; col:9]*	 **Protecting my product and what it's attached to** For example, if the pet that the product is attached to becomes lost, the guardian's contact information can be checked by placing a mobile device on the product to recognise it. **Checking a product's location** The location of the product is based on GPS and Wi-Fi positioning results of the mobile device connected to the product or the nearby Android mobile device, which may have some problems because of things like the surrounding environment or GPS failure.

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		Source: https://downloadcenter.samsung.com/content/UM/202311/20231107134122000/EI-T5600_UM_GLOBAL_TYPE_Rev.1.2_231107.pdf
setting user preferences including geo-fencing parameters, sensitivity of motion sensors, personalized alert output, personalized alert events, or a combination thereof;	"The user preferences include geo-fencing parameters, sensitivity of motion sensors, personalized alert output, personalized alert events, or a combination thereof. The alert includes an audio alarm, an alarm light, an alert message comprising GPS coordinates of the tagged receiver, or a combination thereof." *[I:23-28; col 3]*	**Comment:** *Setting user preferences including geo-fencing parameters, sensitivity of motion sensors, personalized alert output, personalized alert events, or a combination thereof (i.e., geo-fencing is implemented in the Smart Tag 2)* *The Samsung Galaxy SmartTag 2 utilizes geo-fencing through the "Notify when left behind" feature, enabling users to receive alerts when the tag deviates from a predefined range. Personalized alert output is supported via options to customize ringtones and adjust volume levels in the SmartThings app. Personalized alert events can be configured by enabling/disabling specific notifications, like disconnection alerts.*  Getting notified when the product is left behind You can receive a notification on your mobile device when the product is left behind. 1 Open the **SmartThings** app on your mobile device. 2 Tap **Life** → **Find** → **More** → **Notify when left behind** and tap the switch to turn it on. You can set a safe location to avoid notifications, such as your home, so whenever you leave you will not get a notification. Try adding a specific place by tapping **Add safe place**. Source: https://www.bhphotovideo.com/lit_files/1056749.pdf

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		General settings You can check the information related to the product and change various settings. 1 Open the **SmartThings** app on your mobile device. 2 Tap **Devices** and tap your product card. **Volume** Adjusts the volume of the product. Tap **Volume** to select an option you want. You can also tap ▶ to make the product alert a notification sound. **Ringtone** Change the notification sound that the product emits when you are trying to find the product's location. Tap **Ringtone** to select the notification sound you want. Source: https://downloadcenter.samsung.com/content/UM/202311/20231107134122000/EI-T5600_UM_GLOBAL_TYPE_Rev.1.2_231107.pdf
monitoring the asset for an alert event;	"In blocks 1904 and 1905, the control device 200 monitors a status of the secured asset. For example, the	**Comment:** *monitoring the asset for an alert event (i.e., monitor the status of the Smart Tag 2 from the Smart Things app).* *Once the SmartTag 2 is paired with the user's smartphone, Smart Things app enables the continuous monitoring of the SmartTag 2.*

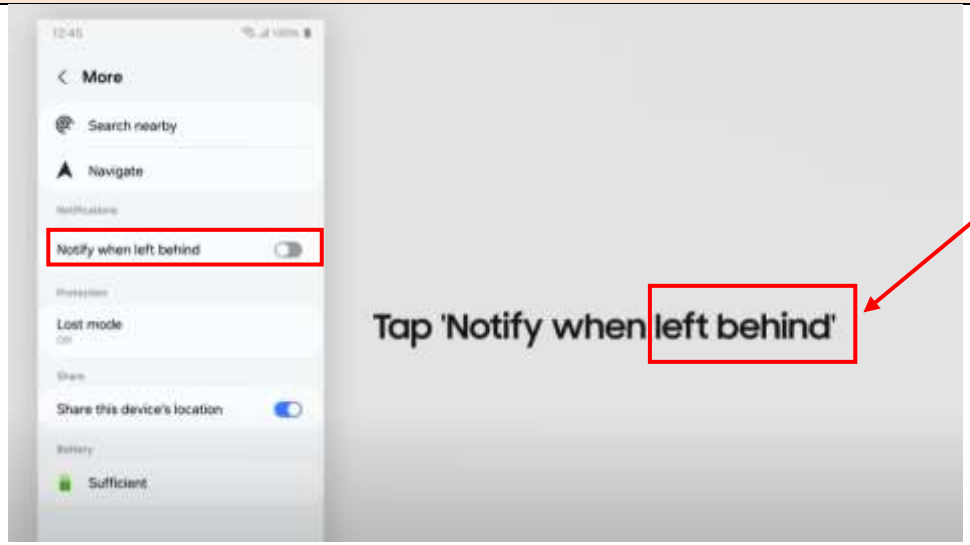
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	control device 200 monitors, using its network interface hardware 218, the tagged receiver 300 for any signals emitted over the network 222 indicative of an alert event. The control device 200 may continuously monitor the status of the device until an alert event occurs. Should an alert event occur, the alarm may be activated, block 1906." *[l:50-57; col 19]*	**Checking a product's location** You can remotely check your product's location from the SmartThings app on your mobile device and find its location by triggering a notification sound on the product. 1 Open the **SmartThings** app on your mobile device. 2 Tap **Devices**, tap your product card, and then tap **View map**. Alternatively, tap **Life** → **Find**. The current location of the product will be displayed on the map. • The location of the product is based on GPS and Wi-Fi positioning results of the mobile device connected to the product or the nearby Android mobile device, which may have some problems because of things like the surrounding environment or GPS failure. • If the current location of the product does not appear, the product's battery has been discharged or there is no Samsung Android mobile devices nearby to help locate the product. To receive notifications when your product's location information is updated, turn on the related feature. Refer to Notify me when it's found for more information. • When the product's status is **Connected**, you can trigger a notification sound or adjust the volume of the product. • The product's usage time may vary depending on the user's environment and the usage pattern. • The Ultra-Wideband (UWB) feature may not be available depending on the region. Source: https://www.bhphotovideo.com/lit_files/1056749.pdf **General settings**

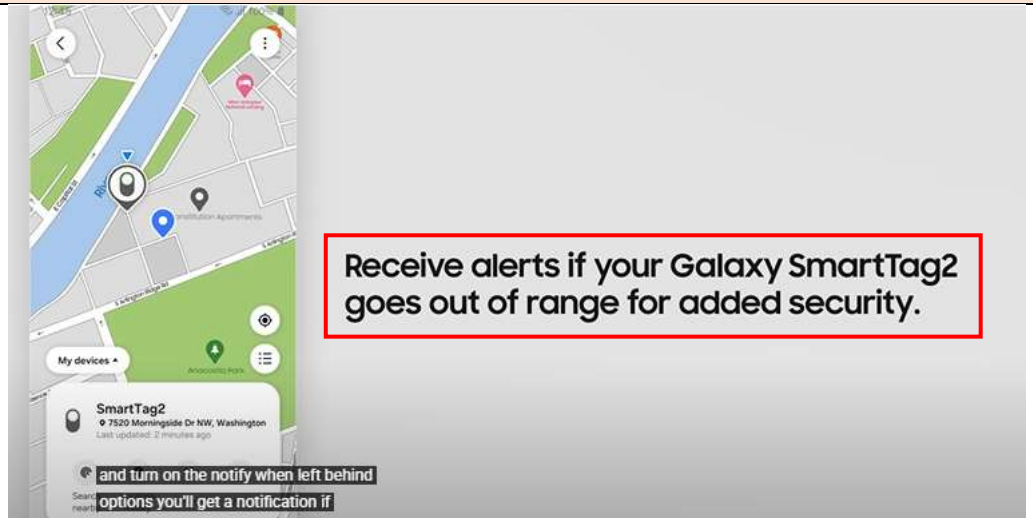
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		Status You can check the connection status between the product and the mobile device and the location status of the product. Tap **View map** to check the detailed location information of the product. **Checking a product's location** You can remotely check your product's location from the SmartThings app on your mobile device and find its location by triggering a notification sound on the product. 1 Open the **SmartThings** app on your mobile device. 2 Tap **Devices**, tap your product card, and then tap **View map**. Alternatively, tap **Life** → **Find**. The current location of the product will be displayed on the map. • The location of the product is based on GPS and Wi-Fi positioning results of the mobile device connected to the product or the nearby Android mobile device, which may have some problems because of things like the surrounding environment or GPS failure. Source: https://downloadcenter.samsung.com/content/UM/202311/20231107134122000/EI-T5600_UM_GLOBAL_TYPE_Rev.1.2_231107.pdf

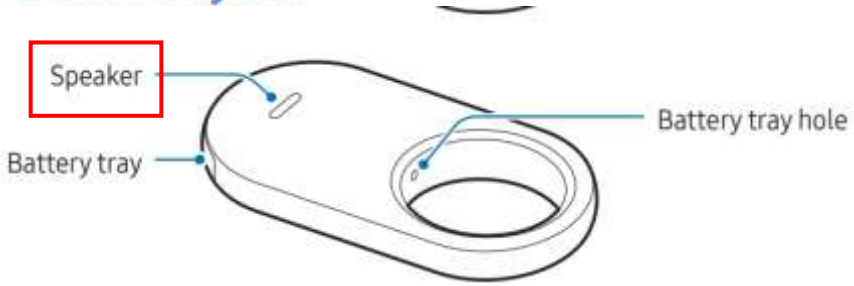
US9786153B2	Specification Support	Galaxy SmartTag2
		 Source: https://www.youtube.com/watch?v=YLm-tUMy2qU
determining that the alert event has occurred;	"In blocks 1904 and 1905, the control device 200 monitors a status of the secured asset. For example, the control device 200 monitors, using its network interface hardware 218, the tagged receiver 300 for	**Comment:** *determining that the alert event has occurred (i.e., notify when left behind).* *The system continuously evaluates the status of the SmartTag 2 relative to predefined conditions, such as proximity to the user's smartphone. For example, when the "Notify when left behind" feature is enabled, the system actively determines whether the SmartTag 2 has moved beyond the user-defined range or has been disconnected from the control device. Upon detecting such a condition, the system identifies it as an alert event and proceeds to trigger a corresponding response.* Use the Samsung Find app notification feature

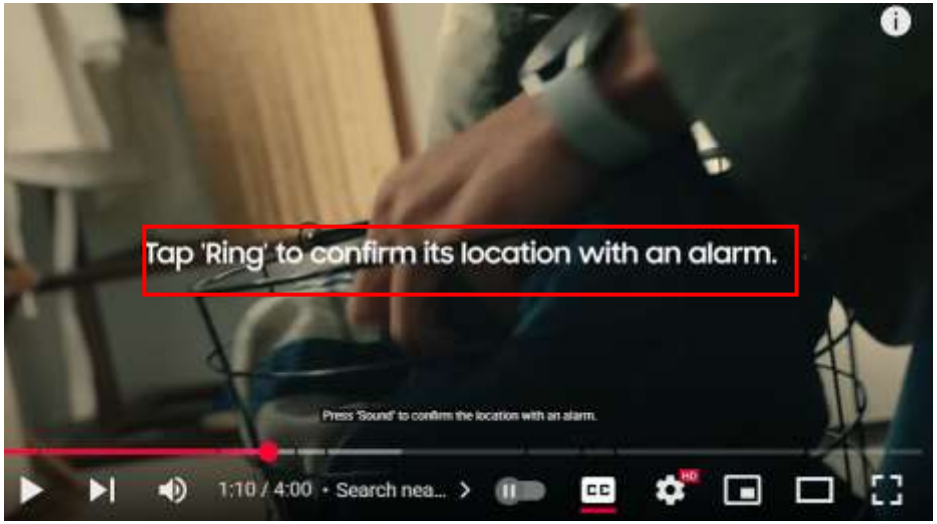
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	any signals emitted over the network 222 indicative of an alert event. The control device 200 may continuously monitor the status of the device until an alert event occurs. Should an alert event occur, the alarm may be activated, block 1906." *[l:50-57; col 19]*	Getting notified when the product is left behind You can receive a notification on your mobile device when the product is left behind. 1 Open the **SmartThings** app on your mobile device. 2 Tap **Life** → **Find** → **More** → **Notify when left behind** and tap the switch to turn it on. Source: https://downloadcenter.samsung.com/content/UM/202311/20231107134122000/EI-T5600_UM_GLOBAL_TYPE_Rev.1.2_231107.pdf **Samsung Find's feature, Notify when left behind, is designed to help users avoid losing important items, like earbuds. The feature sends notifications to the user when they are far enough away from the item that the connection between the item and the user's phone is broken.** Source: https://www.samsung.com/levant/support/apps-services/use-the-samsung-find-app-notification-feature
activating an alert in response to determining that the alert	*"As noted above, the tagged receiver 300 comprises the speaker 322 for*	**Comment:** *activating an alert in response to determining that the alert event has occurred (i.e., notify when left behind), wherein the alert includes a user programmable audio alarm output with a speaker (i.e., ring through Smart Tag 2), wherein the user programmable audio alarm (i.e., customize the ringtone) optionally comprises a high decibel alarm (i.e., high-volume settings) that is output at a higher decibel than an output of the speaker when the alert event has not occurred.*

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event has occurred, wherein the alert includes a user programmable audio alarm output with a speaker, wherein the user programmable audio alarm optionally comprises a high decibel alarm that is output at a higher decibel than an output of the speaker when the alert event has not occurred; and	*transforming data signals from the tagged receiver 300 and/or the control device 200 into mechanical vibrations, in order to output audible prompts or audible information from the tagged receiver 300. The speaker 322 is coupled to the communication path 304 and communicatively coupled to the one or more processors 302. When actuated by the one or more processors 302, the speaker 322 may output an alert indicating that the asset to*	*Upon detecting an alert condition such as disconnection from the control device or movement beyond a defined range the system activates an alert by triggering the built-in speaker on the SmartTag 2. Through the SmartThings app, users are able to customize the ringtone and adjust the volume thereby supporting the requirement for a user-programmable audio alarm. Additionally, third-party sources and product documentation indicate that the tag's high-volume setting can reach up to approximately 105 dB, which is significantly louder. This supports with the claim's requirement for an alarm that outputs at a higher decibel.* **How does SmartThings Find work** **SmartThings Find** has features such as tracking lost devices, activating ringing sounds, locking devices, deleting/backing up data to help you find and manage your devices and protect your information. If you have more than one device, you can also check them through **SmartThings Find** on other devices. Source: https://www.samsung.com/levant/support/apps-services/what-is-find-my-mobile/

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	which the tagged receiver 300 is attached is either being stolen or tampered with. The speaker may comprise at least a 100 db output." [I:8-19; col: 14]	

US9786153B2	Specification Support	Galaxy SmartTag2
		 Source: https://www.youtube.com/watch?v=YLm-tUMy2qU&t=146s **Ringtone** Change the notification sound that the product emits when you are trying to find the product's location. Tap **Ringtone** to select the notification sound you want. **Volume** Adjusts the volume of the product. Tap **Volume** to select an option you want. You can also tap ▶ to make the product alert a notification sound. Source: https://www.youtube.com/watch?v=YLm-tUMy2qU&t=146s

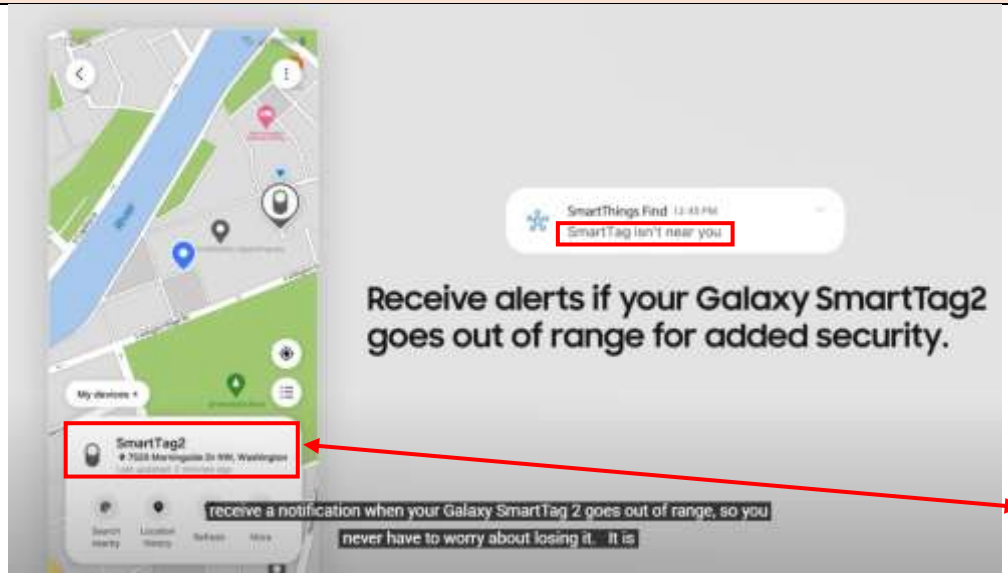
US9786153B2	Specification Support	Galaxy SmartTag2
		Device layout  • If the product has been immersed in water or speaker is wet, sound may not be heard clearly. Source: https://downloadcenter.samsung.com/content/UM/202311/20231107134122000/EI-T5600_UM_GLOBAL_TYPE_Rev.1.2_231107.pdf There's a speaker inside the SmartTag 2 which sends out a high-pitched ringtone when you're trying to find it. You can select the volume level from within the app and even choose different ringtones. There's also a feature that notifies you when the tag is left behind. It's useful if you have a habit of forgetting your keys, for example. As soon as your phones goes out of range, it'll give you a notification that you're leaving the SmartTag 2 behind so you don't forget to take whatever it's attached to along with you. Source: https://r2.community.samsung.com/t5/Wearables/Galaxy-SmartTag-2-review-What-every-Samsung-phone-owner-needs/td-p/15484723

US9786153B2	Specification Support	Galaxy SmartTag2
		 Source: https://www.youtube.com/watch?v=YLm-tUMy2qU **Samsung Galaxy SmartTag2, Bluetooth Tracker, Locator Tracking Device, Item Finder for Keys, Wallet, Luggage, Use w/Phones Tablets Android 11 or Later, 2023, 4 Pack, 2 Black, 2 White**

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		★★★★★ They work just as promised, and great price! Color: Black/White Great tags and work as promised! Very easy to set up. All I had to do was pull the blue tab out, a pairing request popped up on my phone and opened smartthings. It also notified of an update. You can change the ringtone and volume in the app as well. Its pretty loud, 105 db, which is about the decibe Source: https://www.shopabunda.com/products/samsung-galaxy-smarttag2-bluetooth-tracker-locator-tracking-device-item-finder-for-keys-wallet-luggage-use-w-phones-tablets-android-11-or-later-2023-4-pack-2-black-2-white
transmitting an alert message comprising GPS coordinates of the tagged receiver to the control device, an internet server, or a combination thereof	*"During the alert the tagged receiver 300 will output the audio alarm as described herein (block 2002), cause the alarm lit indicator to intensify and or strobe (block 1803), transmit a message sent of the event to the user and or the police (block 2004), transmit GPS coordinates*	**Comments:** *transmitting an alert message (i.e., alert notification) comprising GPS coordinates (i.e., location coordinates) of the tagged receiver (i.e., Samsung Smart Tag 2) to the control device, an internet server, or a combination thereof.* *When the Samsung Galaxy SmartTag 2 is out of range or lost, the SmartThings Find platform, through Samsung's Galaxy Find Network, uses nearby Galaxy devices to detect the tag and estimate its location. Further, the system sends the location data to the user mobile phone, where the user can view the last location of the Smart Tag 2 via Maps and navigate to the location. Since, the Maps requires location coordinates to display any location, it can be inferred that location coordinates are transferred to the control device or mobile device.* Checking a product's location You can remotely check your product's location from the SmartThings app on your mobile device and find its location by triggering a notification sound on the product. 1 Open the **SmartThings** app on your mobile device. 2 Tap **Devices**, tap your product card, and then tap **View map**. Alternatively, tap **Life** → **Find**. The current location of the product will be displayed on the map.

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	to the user and or the police (block 2005).” [l: 66-67; col:19; l:1-5; col:20] *“For example, a user may ask, “Where are my keys?” and the control device 200 may respond through speech output through the speaker 221 and can state the location of the keys in addition to displaying their location (e.g. on a map like Google Earth/Maps or by displaying GPS coordinates, or the like).”[l:44-50; col : 8].*	• The location of the product is based on GPS and Wi-Fi positioning results of the mobile device connected to the product or the nearby Android mobile device, which may have some problems because of things like the surrounding environment or GPS failure. **Getting route guidance** If your product is lost or far away from you, you may be directed to the location where the product is located. 1 Open the **SmartThings** app on your mobile device. 2 Tap **Life** → **Find** → **More** → **Navigate**. Directions to the place where the product is located will begin. If a supported navigation app is not installed on your mobile device, the list of supported navigation apps will appear. Install the app you want from the list and get directions. Source: https://downloadcenter.samsung.com/content/UM/202311/20231107134122000/EI-T5600_UM_GLOBAL_TYPE_Rev.1.2_231107.pdf

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Source: https://www.youtube.com/watch?v=YLm-tUMy2qU

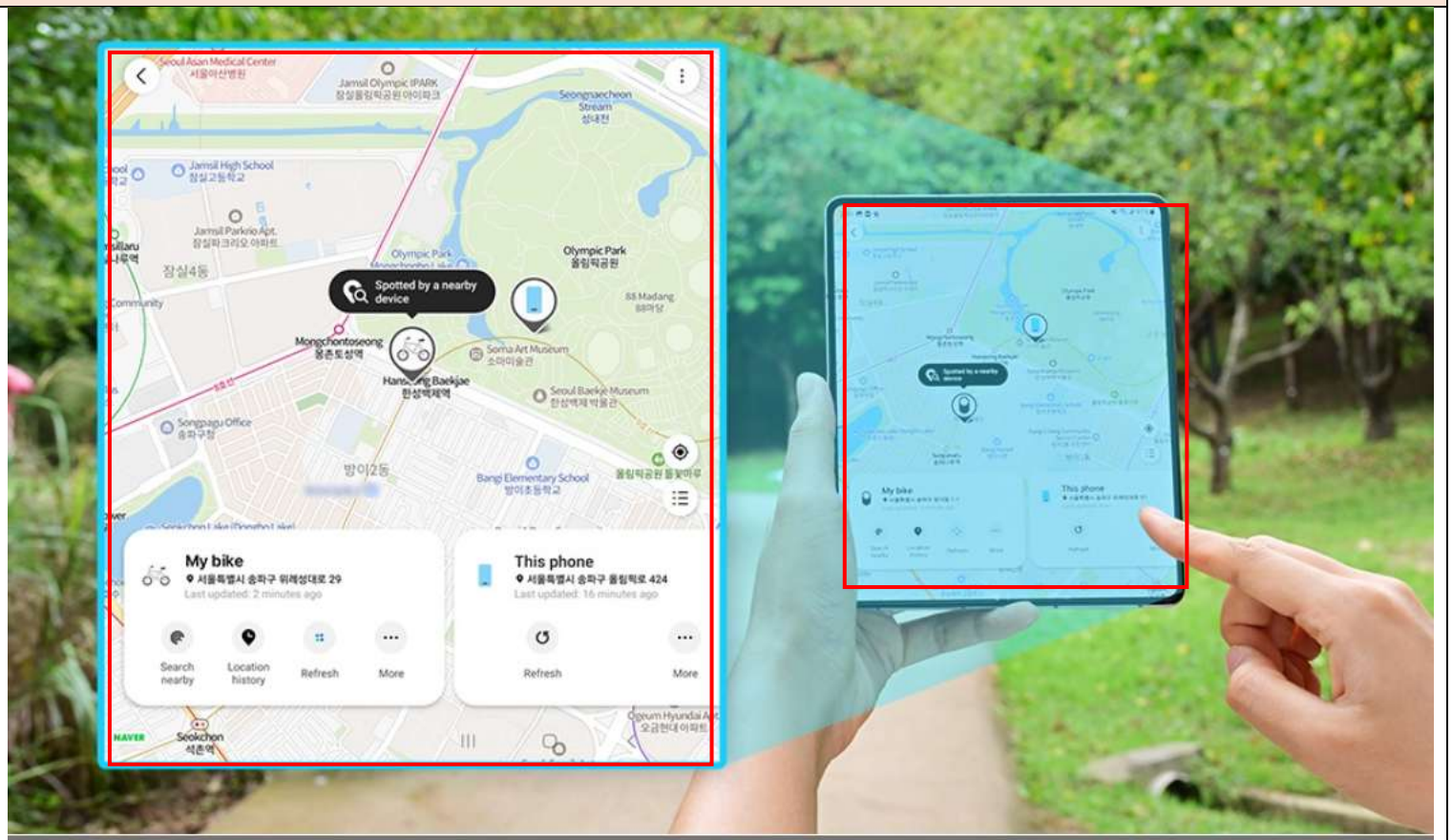
The Offline finding feature in SmartThings Find can help you locate a missing device, even when it's turned off or not online. Once your device has been offline for 30 minutes, it will produce a signal that can be received by other devices. A nearby Galaxy device user can then assist you with finding your lost device by sending that information to Samsung's server. On the other hand, the feature also allows you to encrypt your device's offline location so only you can access it. However, you'll first need to turn on the Offline finding feature in order for the service or offline encryption to work.

Source: https://www.samsung.com/us/support/answer/ANS10004549/

SmartThings Developers

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		getAlertLinkByLocationId Redirect to condent provider's alert detail link. If no query given, coordinate of the location ID is used. Otherwise the coordinate of subscription will be used. Source: https://www.youtube.com/watch?v=YLm-tUMy2qU

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		## Checking a product's location You can remotely check your product's location from the SmartThings app on your mobile device and find its location by triggering a notification sound on the product. 1 Open the **SmartThings** app on your mobile device. 2 Tap **Devices**, tap your product card, and then tap **View map**. Alternatively, tap **Life** → **Find**. The current location of the product will be displayed on the map. • The location of the product is based on GPS and Wi-Fi positioning results of the mobile device connected to the product or the nearby Android mobile device, which may have some problems because of things like the surrounding environment or GPS failure. • If the current location of the product does not appear, the product's battery has been discharged or there is no Samsung Android mobile devices nearby to help locate the product. To receive notifications when your product's location information is updated, turn on the related feature. Refer to Notify me when it's found for more information. • When the product's status is **Connected**, you can trigger a notification sound or adjust the volume of the product. • The product's usage time may vary depending on the user's environment and the usage pattern. • The Ultra-Wideband (UWB) feature may not be available depending on the region. Source: https://www.bhphotovideo.com/lit_files/1056749.pdf

US9786153B2	Specification Support	Galaxy SmartTag2
		 ▲ You can check the location of your bike with SmartThings Find. Source: https://news.samsung.com/global/user-guide-guard-your-bike-gear-with-the-galaxy-smarttag2

Disclaimer

This claim chart is prepared for informational and analysis purposes only. It is not intended to allege infringement by any party but to compare the publicly available information of Samsung with the claims of the referenced patent(s). The content herein does not constitute a legal opinion or an assertion of liability. Any conclusions drawn from this chart should be independently verified.